



Forensic dentistry



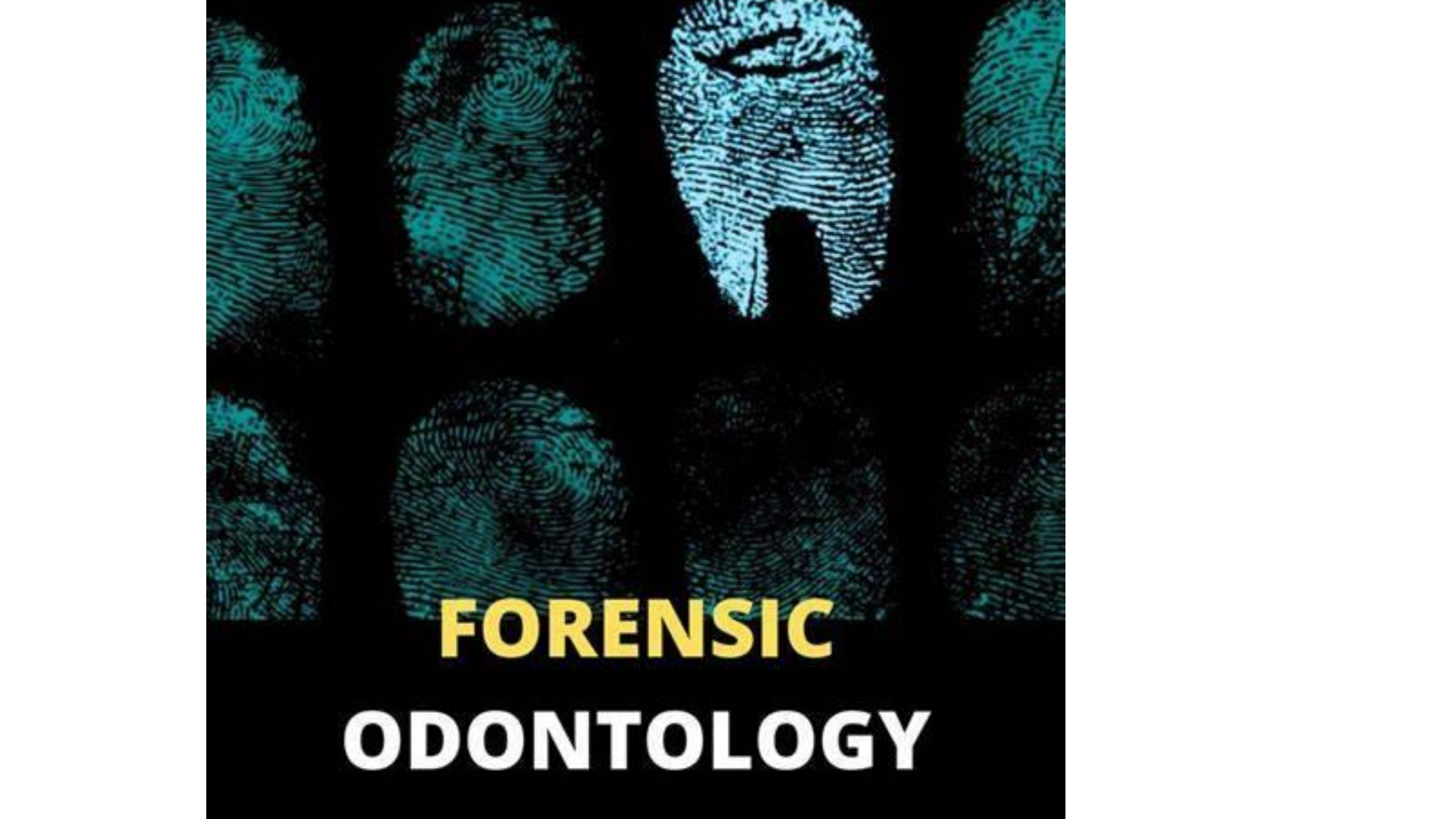
Introduction

Forensic dentistry or forensic odontology is the proper handling, examination and evaluation of dental evidence, which will be then presented in the interest of justice.



The teeth and dental restorations are the strongest elements in the human body and survive the destructive influences of fire and exposure to the elements. Forensic identification plays a major role in man-made or natural disaster.



The background of the slide is a dark, textured surface featuring several fingerprints. The fingerprints are rendered in a light blue or cyan color, creating a high-contrast, forensic aesthetic. The ridges and valleys of the fingerprints are clearly visible, and they are arranged in a somewhat grid-like pattern across the upper and middle portions of the slide.

FORENSIC ODONTOLOGY

In the case of forensic dentistry, experts (forensic dentists) can use dental records for:

- I. Identification of found human remains.
- II. Identification the suspect



Identification of found human remains:

It was done by using dental records, the principle of dental identification is that:

- Postmortem dental remains can be compared with antemortem dental records, including written notes, study casts, radiographs, photographs etc, to confirm identity.
- If there is no antemortem dental records, a postmortem dental profile will typically provide information on the victim's:

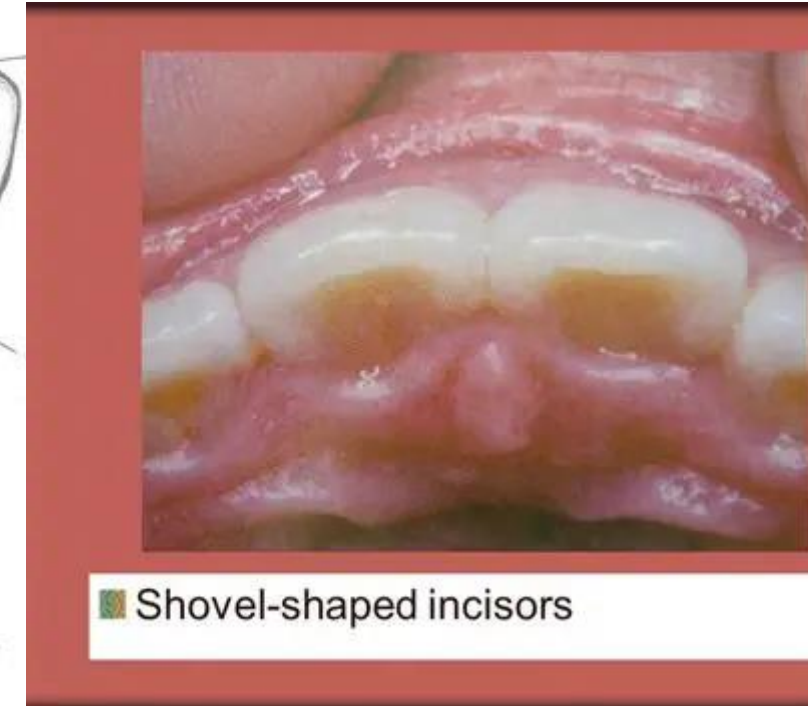
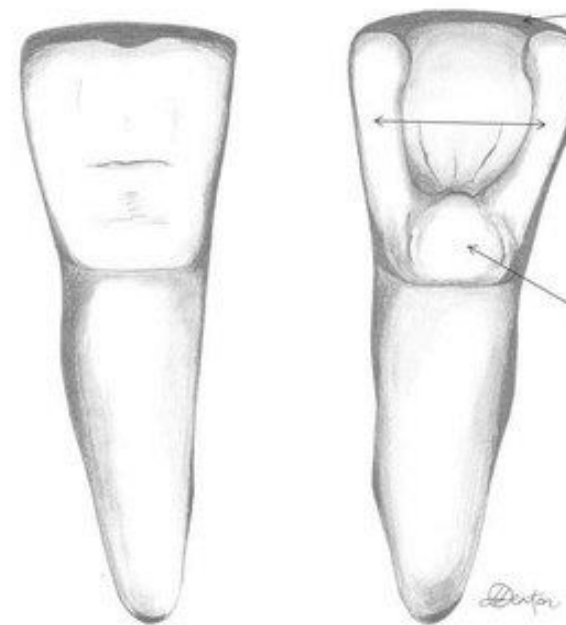


1–Age:

- In children: The patterns of tooth eruption, the root length were assessed.
- In young adults: The third molar development.
- In middle-aged and older adults: Periodontal disease progression, excessive wear, multiple restorations, extractions were assessed.

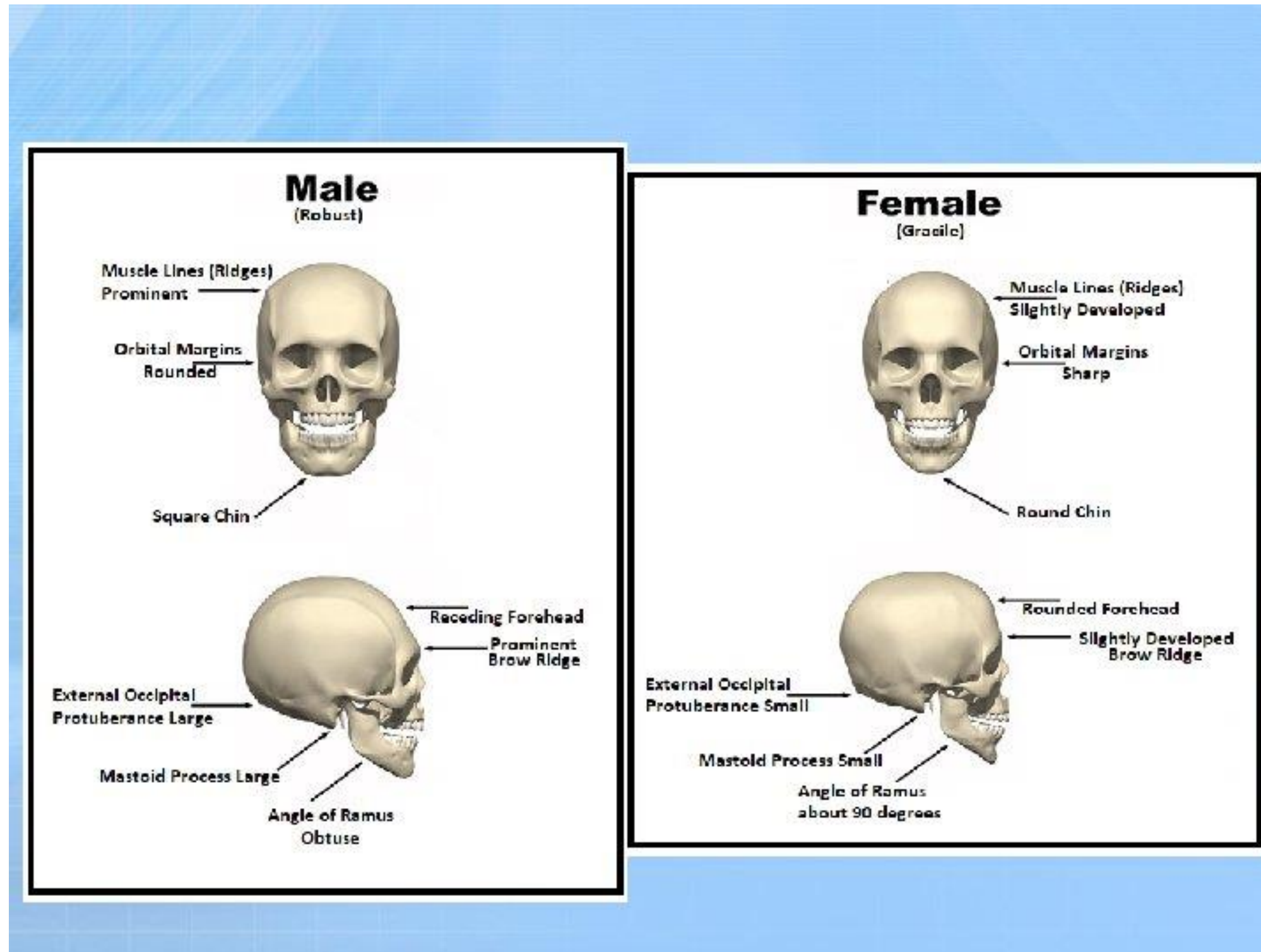
2–Race

Can be assessed from skull shape and form additional characteristics, such as cusps of Carabelli, shovel-shaped incisors.



3-Gender can be assessed from

- Skull shape and form, (no gender differences regarding teeth morphology).
- Presence or absence of Y-chromatin in teeth.
- DNA analysis.
- Mandibular canines' size.



4– Socio-economic status

can be assessed through the quality, quantity and presence or absence of dental treatment.

5–Occupation, dietary habits and dental or systemic diseases.

The presence of erosion can suggest alcohol or an eating disorder while stains can indicate smoking, tetracycline. Unusual wear patterns may result from pipe stems, cigarette holders.

II. Identification the suspect:

Identification the suspect through the assessment of bite mark injuries in cases of abuse in (child, elder) and in women during sexual attacks

Bite marks can be found on:

- The victim (by the attacker).
- The attacker (suspect) when a victim attempts to defend himself.
- An object found at the crime scene.

Evidence collection from the bite victim: Dentists should be familiar with the general principles of evidence collection. These are:

- 1. Documentation:** Make a descriptive record of the injury, including the physical appearance, colour, size and orientation of the injury, location on the body, relative contour and elasticity of the site, and types of injuries.
- 2. Photographs:** Take photographs, both colour and black-and-white films.
- 3. Saliva swab:**
 - a. Saliva will have been deposited on the skin during biting and this should be collected and analyzed.
 - b. sample of whole blood must be collected from the victim at this time to assess the victim's DNA
- 4. Impression:** Fabricate an impression of the bitten surface to record any irregularities produced by the teeth

Bite Mark Photography:



Matching mold to bite mark



Evidence collection from the bite suspect: The following evidence are recovered during examination of the bitemark suspect:

1. **Clinical examination** The extra and intra-oral structures are examined and are noted on a dental chart. Special attention is focused on the status of the dental health, occlusion and mandibular articulation, tooth mobility, periodontal pocketing, dental restorations, diastemata, fractures, caries, etc., and the function of masticatory muscles.
2. **Photographs** Full facial and profile photographs are produced in addition to frontal and lateral views of the teeth in occlusion.
3. **Impressions** It is necessary to produce extremely accurate study casts of the teeth that record all characteristics of the dentition.
4. **Bite sample** A sample of the suspect's bite is recorded in centric occlusion using a wax.
5. **Salivary sample** Saliva is also taken for DNA testing.

The suspect's DNA profile obtained from saliva or blood with salivary DNA surrounding the bite mark area proves to be a more reliable form of identification.



DNA Analysis

Because of the resistant nature of dental tissues to environmental assaults, such as incineration, immersion, trauma, mutilation and decomposition, teeth represent an excellent source of DNA material. DNA analysis, also known as DNA fingerprinting, is a fairly new technology that may replace dental identification and fingerprint identification as the most definitive means of identifying unknown remains. Dental DNA analysis (the coronal pulp chamber and radicular canal are obvious target for DNA sampling).



**THANKS
FOR LISTENING**

